Linear Algebra I & Mathematics Tutorial 1b Instructor: Henrik Bachmann
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Quiz 4: Linear maps

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Exercise 2. (4 Points) Let $F : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear map with

$$F\begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}3\\5\\-2\end{pmatrix}, \qquad F\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1\\4\\-2\end{pmatrix}.$$

Calculate $F\begin{pmatrix}3\\1\end{pmatrix}$.

Solution.

$$\begin{bmatrix}3\\1\end{bmatrix} = \overset{\cup}{\cup}\begin{bmatrix}2\\1\end{bmatrix} + \overset{\cup}{\sim}\begin{bmatrix}1\\1\end{bmatrix} \qquad \begin{matrix}4\\2\end{matrix} - \begin{matrix}1\\1\end{matrix} = \begin{matrix}3\\1\end{matrix}$$

represented by:

$$\overset{\to}{\underset{-1}{\circlearrowleft}}\left[\begin{array}{cc|c}2&1&3\\1&1&1\end{array}\right] \sim \overset{-1}{\underset{\to}{\circlearrowleft}}\left[\begin{array}{cc|c}1&0&2\\1&1&1\end{array}\right] \sim \left[\begin{array}{cc|c}1&0&2\\0&1&-1\end{array}\right]$$

$$\overset{\cup}{\cup} = 2$$
$$\overset{\cup}{\sim} = -1$$

so: because F is linear,

$$F\left(\begin{bmatrix}3\\1\end{bmatrix}\right) = F\left(2\begin{bmatrix}2\\1\end{bmatrix} - \begin{bmatrix}1\\1\end{bmatrix}\right) = 2F\left(\begin{bmatrix}2\\1\end{bmatrix}\right) - F\left(\begin{bmatrix}1\\1\end{bmatrix}\right) = 2\begin{bmatrix}3\\5\\-2\end{bmatrix} - \begin{bmatrix}1\\4\\-2\end{bmatrix} = \boxed{\begin{bmatrix}5\\6\\-2\end{bmatrix}}$$

$$F\left(\begin{bmatrix}3\\1\end{bmatrix}\right) = \begin{bmatrix}5\\6\\-2\end{bmatrix}$$